

CSA3006 - DATA MINING AND DATA WAREHOUSING

Presented By

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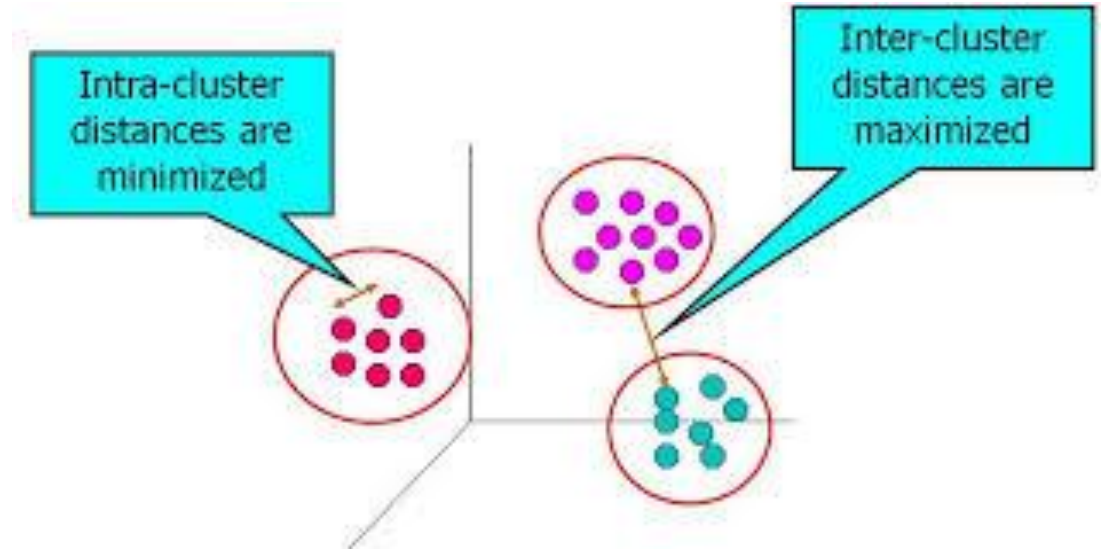
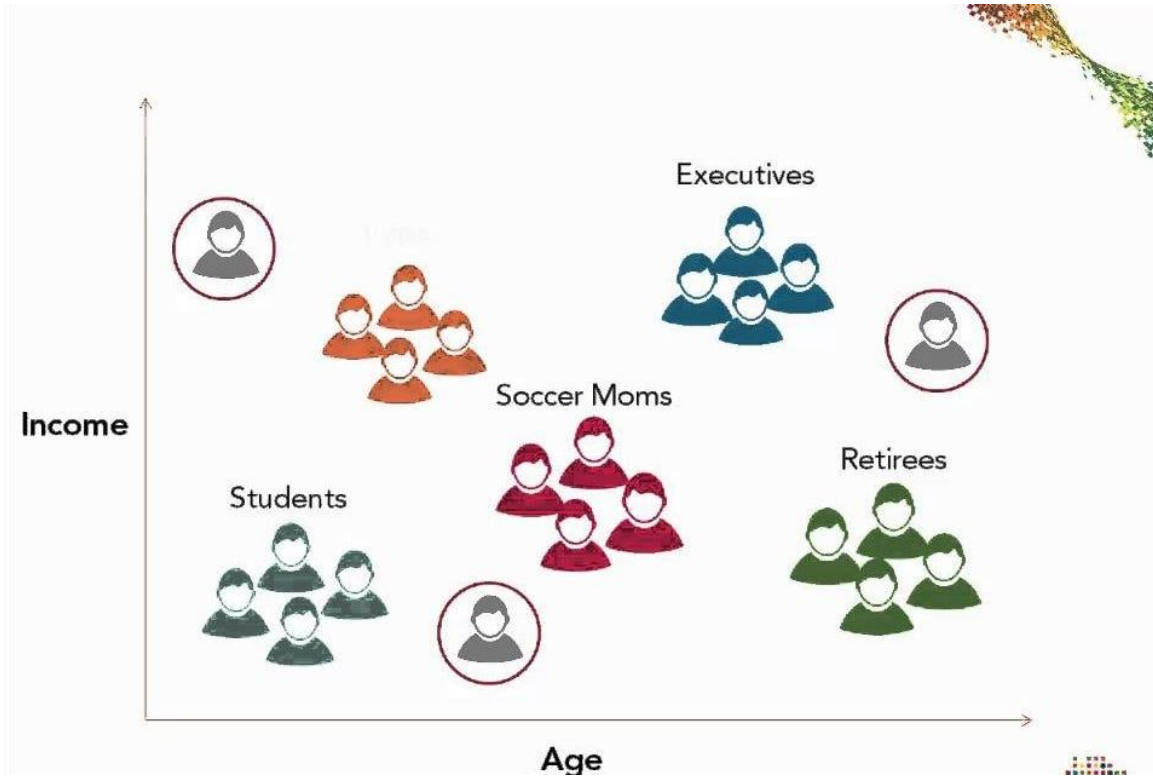
VIT Bhopal University

Module IV

- **Clustering and Applications:** Basic concepts (cluster, centroid), Cluster analysis—Types of Data in Cluster Analysis—Categorization of Major Clustering Methods— algorithms (KMeans, Fuzzy CMeans), Hierarchical Methods— Density—Based Methods, Statistical-based clustering (EM algorithm). Graph—Based Methods, Spectral techniques Outlier Analysis.

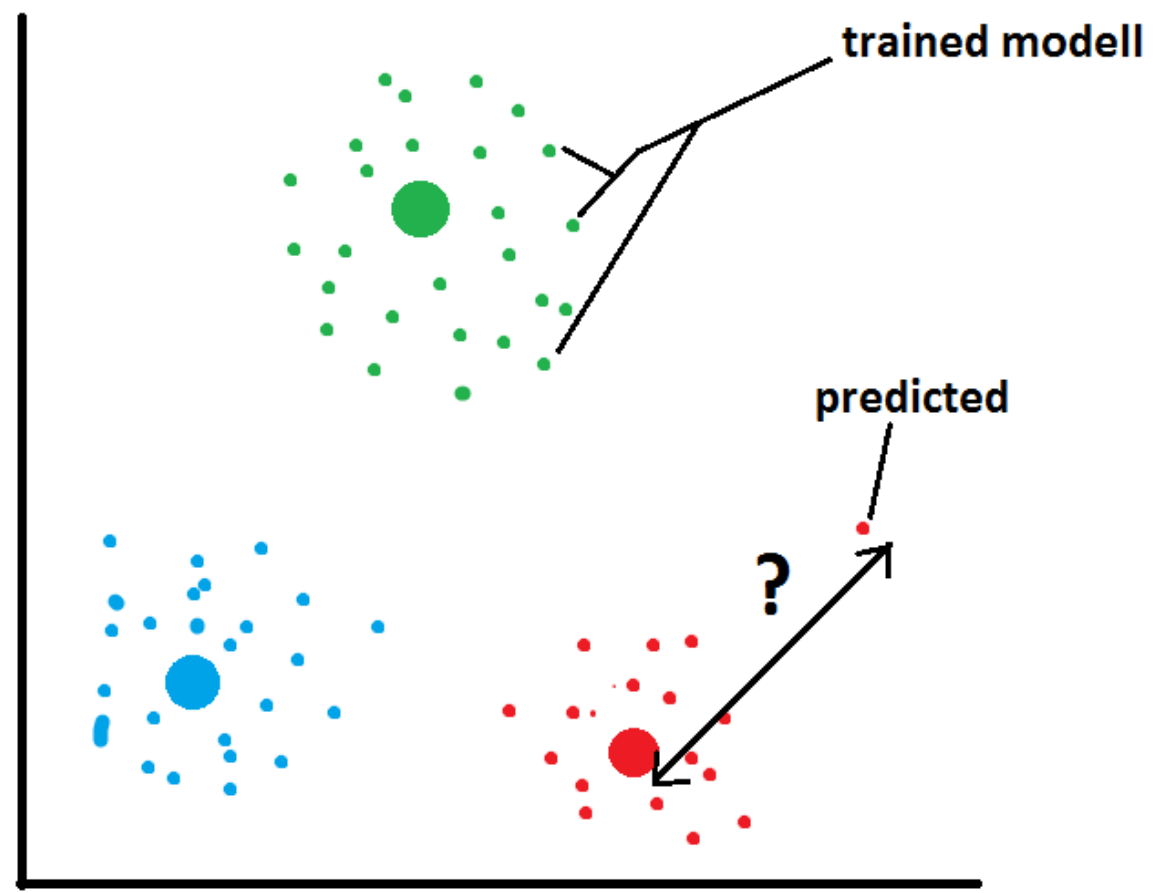
Basic Concepts - Cluster

- A **cluster** is a collection of data objects that are:
- **Similar to each other** within the same group
- **Dissimilar to objects** in other groups
- It is the **result of clustering process**
- Represents a **group of homogeneous data**
- Used for **data compression** (treat group as one unit)
- Helps in identifying **hidden patterns in data**
- Also called **data segmentation**
- **Characteristics:**
- High **intra-cluster** similarity
- Low **inter-cluster** similarity
- **Example:**
- Customers grouped based on purchasing behavior
- Students grouped based on performance



Basic Concepts - Centroid

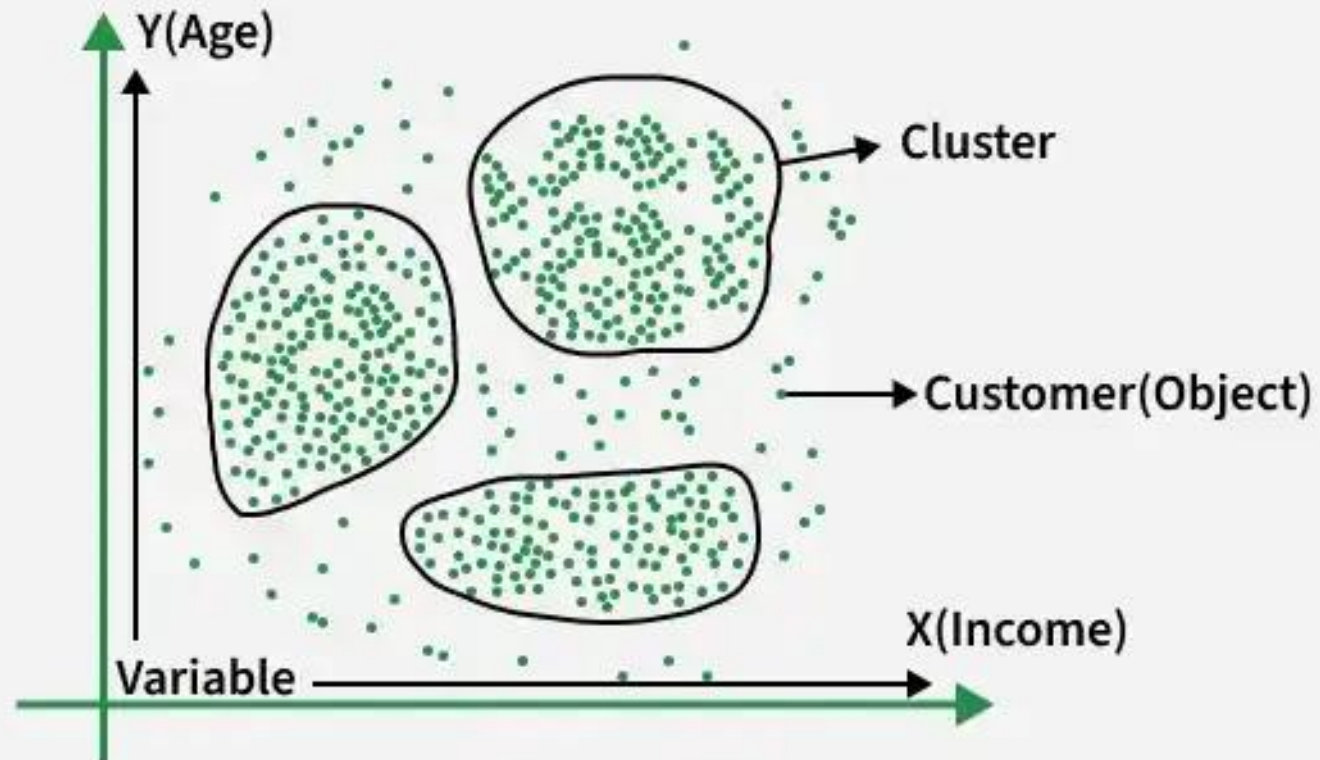
- A **centroid** is the **center point of a cluster**, calculated as the **mean of all data points** in that cluster.
- Represents the **average position of all objects** in a cluster
- Used in **K-Means algorithm**
- Not necessarily an actual data point
- Acts as a **reference point** for assigning objects
- **Mathematical Idea:**
- Centroid = mean of all points in the cluster
- **Role in Clustering:**
 - Assign objects to nearest cluster
 - Update cluster position during iterations
- Minimizes **distance between points and center**
- **Example:**
- In a group of students, centroid = average marks
- In location data, centroid = central location



No	Cluster	Centroid
1	Group of similar data objects	Center point of a cluster
2	Collection of multiple points	Single representative point
3	Groups similar data	Represents the cluster
4	Contains many objects	Derived from objects
5	Always real data points	May not be a real data point
6	Final output of clustering	Helps in forming clusters
7	Based on similarity	Mean of all points
8	Group of similar students	Average marks of students

Cluster Analysis

Cluster Analysis



Requirements of Clustering in Data Mining

A good clustering algorithm must satisfy several real-world requirements:

- **Scalability:** Should handle millions of data points efficiently
- **Different Data Types:** Must support numerical, categorical, and mixed data
- **Arbitrary Shape:** Should detect clusters of any shape (not just circular)
- **Minimal Domain Knowledge:** Should not require many input parameters
- **Noise Handling:** Must handle missing or incorrect data
- **Incremental Ability:** Should update clusters when new data arrives
- **High Dimensionality:** Must handle data with many features
- **Constraint-Based:** Should respect user-defined conditions
- **Interpretability:** Results must be understandable

Real-world datasets are **complex and noisy**, so clustering must be robust.

Distance Metrics in Cluster Analysis

- **Euclidean Distance:** It is the most widely used distance metric and finds the straight-line distance between two points.
- **Manhattan Distance:** It measures the distance between two points **based on grid-like path**. It adds the absolute differences between the values.
- **Cosine Similarity:** This method checks the **angle between two points** instead of looking at the distance. It's used in text data to see how similar two documents are.
- **Jaccard Index:** A statistical tool used for comparing the similarity of sample sets. It's **mostly used for yes/no type data** or categories.

Types of Data in Cluster Analysis

In cluster analysis, the type of data plays a crucial role in determining the **similarity or distance between objects**. Different data types require different handling methods and distance measures.

1. Interval-Scaled Variables

2. Binary Variables

3. Nominal Variables

4. Ordinal Variables

5. Ratio-Scaled Variables

6. Mixed-Type Data

1. Interval-Scaled Variables

- Continuous numeric data measured on a linear scale
- Equal intervals between values
- No true zero point
- **Examples:** Temperature (°C, °F), IQ scores
- **Characteristics:**
 - Supports addition and subtraction
 - Commonly used with **Euclidean distance**

2. It is easy to navigate on forms.app.

1	2	3	4	5
---	---	---	---	---

Strongly agree

Neutral

Strongly disagree

Press **ENTER**

NEXT →

2. Binary Variables

- Data with only two possible values: **0 or 1**
- Represents presence/absence or yes/no

Types:

- **Symmetric Binary:** Both values equally important
- **Asymmetric Binary:** One value (usually 1) is more important

Examples: Gender (M/F), Disease (Yes/No)

8	4	2	1	Dec	Hex
0	0	0	0	0	0
0	0	0	1	1	1
0	0	1	0	2	2
0	0	1	1	3	3
0	1	0	0	4	4
0	1	0	1	5	5
0	1	1	0	6	6
0	1	1	1	7	7
1	0	0	0	8	8
1	0	0	1	9	9
1	0	1	0	10	A
1	0	1	1	11	B
1	1	0	0	12	C
1	1	0	1	13	D
1	1	1	0	14	E
1	1	1	1	15	F

3. Nominal Variables

- Categorical data with **no inherent order**
- Values are just labels
- **Examples:** Color (Red, Blue), Department names
- **Characteristics:**
- Cannot perform arithmetic operations
- Use matching or mismatch-based similarity measures

EXAMPLES OF NOMINAL DATA



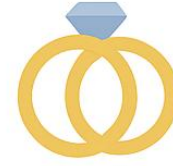
COLORS

Red
Blue
Green
Yellow



BLOOD GROUPS

A
B
AB
O



MARITAL STATUS

Single
Married
Divorced
Widowed



PET TYPE

Dog
Cat
Bird



NATIONALITY

Indian
American
Japanese



MOVIE GENRE

Action
Comedy
Romance

4. Ordinal Variables

- Categorical data with a **meaningful order**
- Ranking is important but exact differences are unknown
- **Examples:** Rank (1st, 2nd, 3rd), Satisfaction (Low, Medium, High)
- **Characteristics:**
- Can be converted into ranks
- Often transformed to numeric values for clustering

Examples of ordinal data



Customer satisfaction

How satisfied are you with our product or service?

- Very satisfied
- Satisfied
- Neutral
- Dissatisfied
- Very dissatisfied



Movie ratings

Rate this movie on a scale of 1 to 5 stars.

- Five stars (amazing)
- Four stars (good)
- Three stars (average)
- Two stars (OK)
- One stars (not good)



Social media sentiment analysis

Categorize the sentiment of this social media post.

- Positive
- Neutral
- Negative



Education levels

What is your highest level of education?

- High school diploma
- Associate degree
- Bachelor's degree
- Master's degree
- Doctorate



T-shirt sizes

What size T-shirt do you wear?

- Small
- Medium
- Large
- Extra Large



Restaurant experience

Rate the food quality at this restaurant.

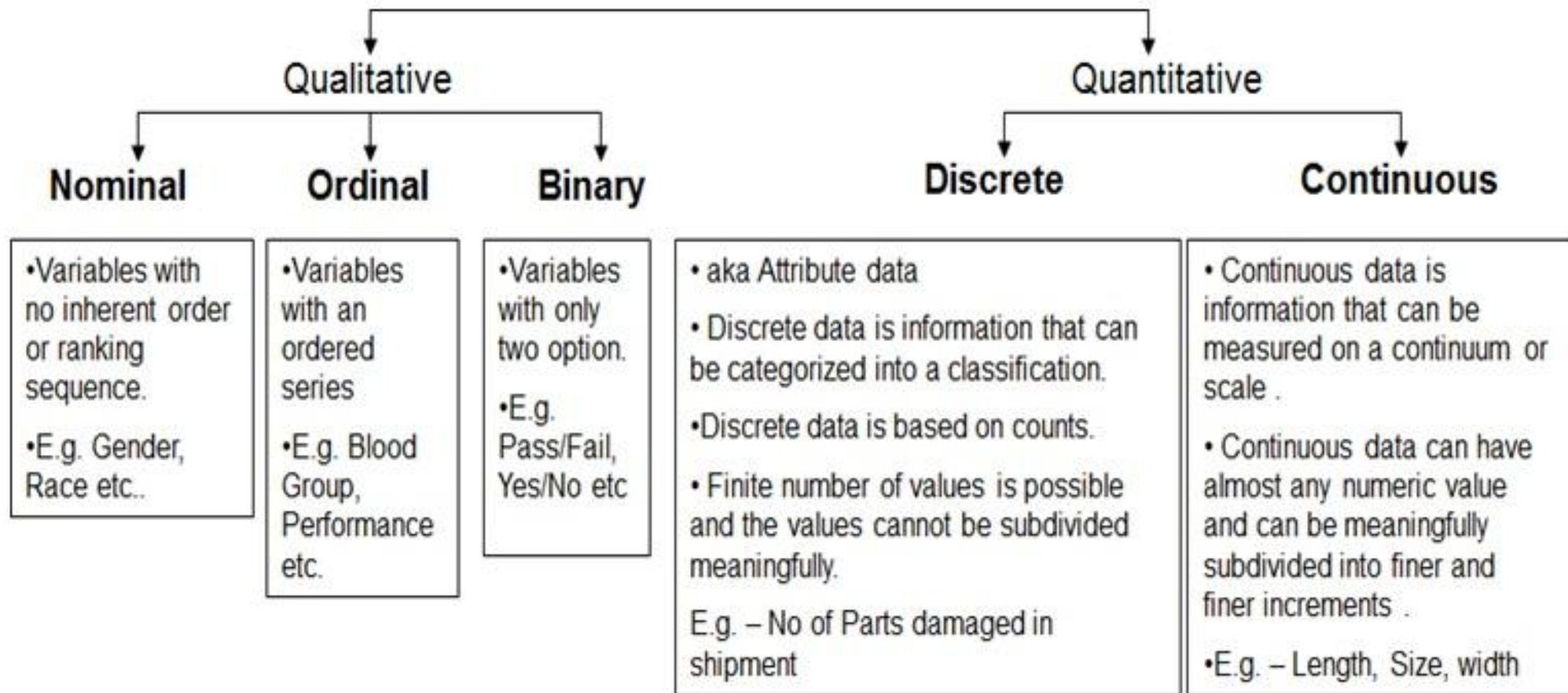
- Excellent
- Very good
- Good
- Average
- Below average

5. Ratio-Scaled Variables

- Numeric data with a **true zero point**
- Ratios are meaningful
- **Examples:** Height, Weight, Age
- **Characteristics:**
- Supports all arithmetic operations
- Widely used in clustering algorithms

6. Mixed-Type Data

- Combination of different data types (e.g., numeric + categorical + binary)
- **Examples:**
 - Patient data (Age, Gender, Disease)
 - Customer data (Income, Region, Purchase behavior)
- **Handling Methods:**
 - Data normalization
 - Use of specialized distance measures (e.g., Gower distance)



Categorization of Major Clustering Methods (Techniques)

Hierarchical Clustering in Data Mining

- Hierarchical clustering groups data into a **tree-like structure of clusters**.
- Initially, **each data point is treated as a separate cluster**.
- The algorithm repeatedly performs two steps:
 - Find the **two closest clusters**.
 - **Merge those clusters** into a single cluster.
- This process continues **until all data points are merged into one cluster**.
- The main goal is to create a **hierarchy of nested clusters**.
- The result is represented using a **dendrogram**.
- A **dendrogram** is a **tree-like diagram** showing how clusters are formed.
- It illustrates the **order of merging (bottom-up)** or **splitting (top-down)** of clusters.

What is Hierarchical Clustering?

- Hierarchical clustering is a method of cluster analysis in data mining that creates a hierarchical representation of the clusters in a dataset.
- The method starts by treating each data point as a separate cluster and then iteratively combines the closest clusters until a stopping criterion is reached.
- The result of hierarchical clustering is a tree-like structure, called a dendrogram, which illustrates the hierarchical relationships among the clusters.

Hierarchical clustering has several advantages over other clustering methods

- The ability to handle non-convex clusters and clusters of different sizes and densities.
- The ability to handle missing data and noisy data.
- The ability to reveal the hierarchical structure of the data, which can be useful for understanding the relationships among the clusters.

Drawbacks of Hierarchical Clustering

- The need for a criterion to stop the clustering process and determine the final number of clusters.
- The computational cost and memory requirements of the method can be high, especially for large datasets.
- The results can be sensitive to the initial conditions, linkage criterion, and distance metric used.
In summary, Hierarchical clustering is a method of data mining that groups similar data points into clusters by creating a hierarchical structure of the clusters.
- This method can handle different types of data and reveal the relationships among the clusters. However, it can have high computational cost and results can be sensitive to some conditions.

Types of Hierarchical Clustering

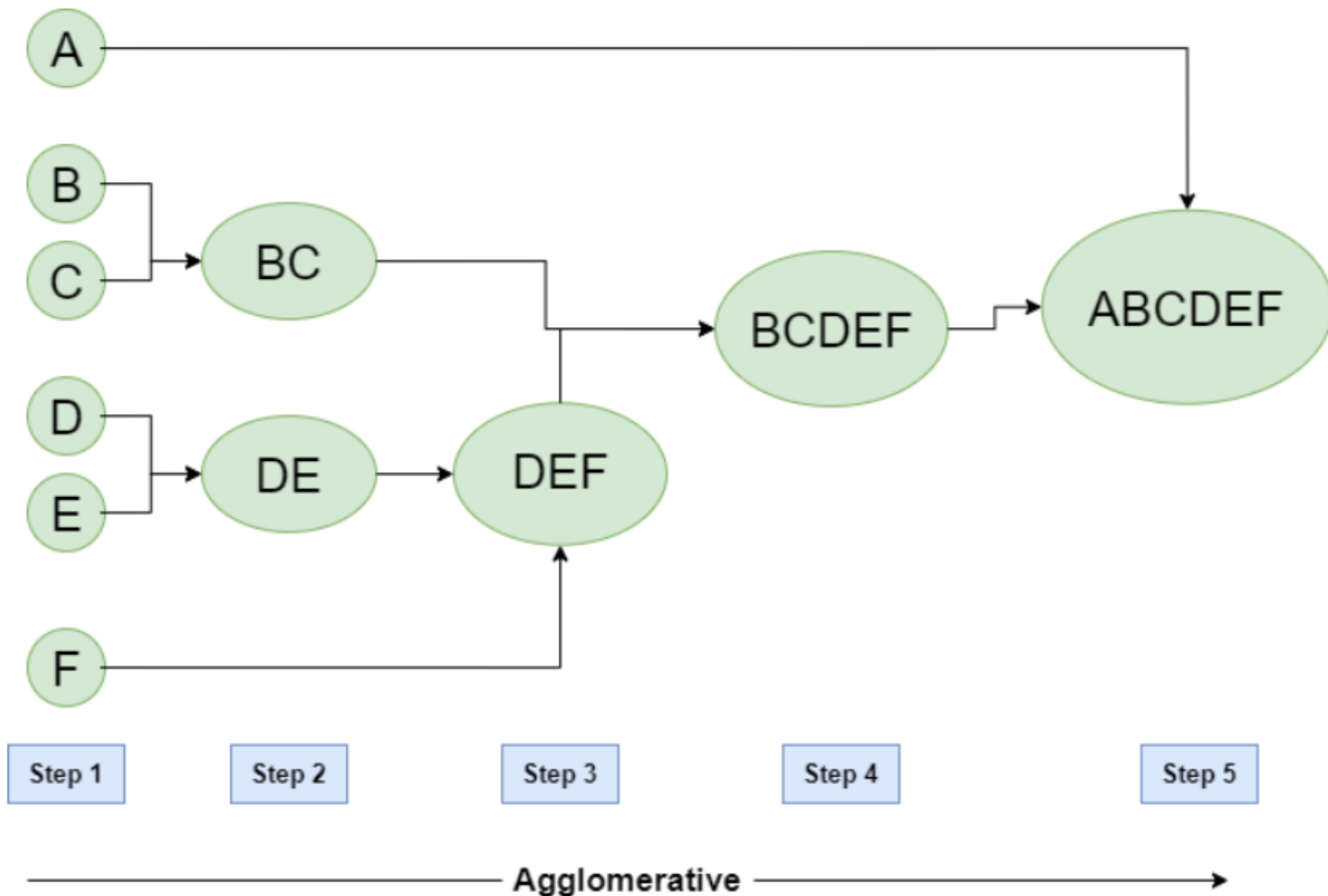
- Basically, there are two types of hierarchical Clustering:
- Agglomerative Clustering
- Divisive clustering

1. Agglomerative Clustering

- Initially consider every data point as an **individual** Cluster and at every step, merge the nearest pairs of the cluster. (It is a bottom-up method).
- At first, every dataset is considered an individual entity or cluster. At every iteration, the clusters merge with different clusters until one cluster is formed.

The algorithm for Agglomerative Hierarchical Clustering is:

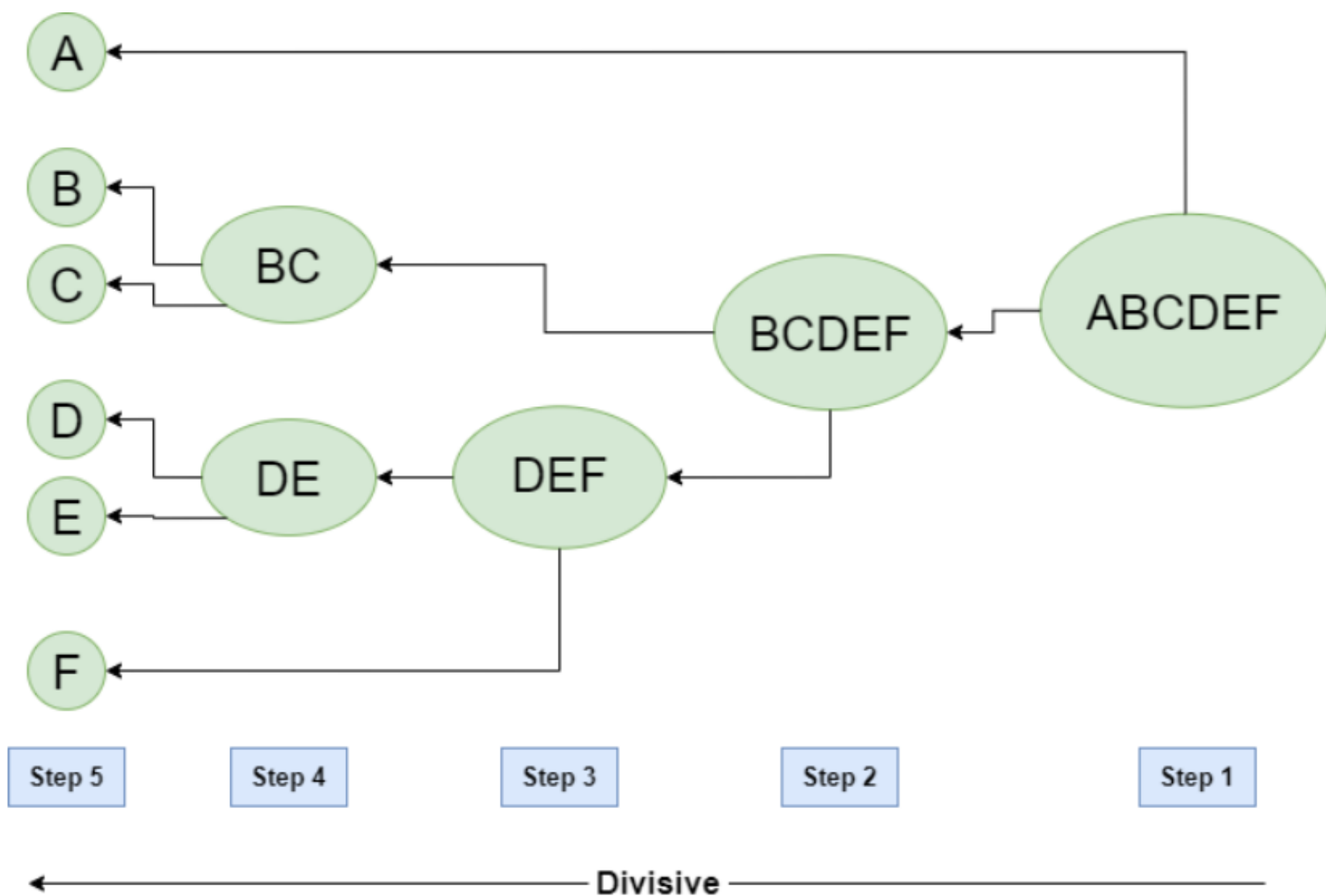
- Calculate the similarity of one cluster with all the other clusters (calculate proximity matrix)
- Consider every data point as an individual cluster
- Merge the clusters which are highly similar or close to each other.
- Recalculate the proximity matrix for each cluster
- Repeat Steps 3 and 4 until only a single cluster remains.



- **Step-1:** Consider each alphabet as a single cluster and calculate the distance of one cluster from all the other clusters.
- **Step-2:** In the second step comparable clusters are merged together to form a single cluster. Let's say cluster (B) and cluster (C) are very similar to each other therefore we merge them in the second step similarly to cluster (D) and (E) and at last, we get the clusters [(A), (BC), (DE), (F)]
- **Step-3:** We recalculate the proximity according to the algorithm and merge the two nearest clusters([(DE), (F)]) together to form new clusters as [(A), (BC), (DEF)]
- **Step-4:** Repeating the same process; The clusters DEF and BC are comparable and merged together to form a new cluster. We're now left with clusters [(A), (BCDEF)].
- **Step-5:** At last, the two remaining clusters are merged together to form a single cluster [(ABCDEF)].

2. Divisive Hierarchical clustering

- We can say that Divisive Hierarchical clustering is precisely the **opposite** of Agglomerative Hierarchical clustering.
- In Divisive Hierarchical clustering, we take into account all of the data points as a single cluster and in every iteration, we separate the data points from the clusters which aren't comparable.
- In the end, we are left with N clusters.



Density Based Clustering

- Here are **simple, clear points** for your content:
- **DBSCAN (Density-Based Spatial Clustering of Applications with Noise)** is a **density-based clustering algorithm**.
- It groups **closely packed data points** into clusters.
- Points in **low-density regions** are marked as **noise (outliers)**.
- Clusters are formed as **dense regions separated by sparse areas**.
- Unlike K-Means or hierarchical clustering, DBSCAN **does not assume clusters are spherical**.
- It works well with **real-world data that has irregular patterns** such as
 - **Arbitrary-shaped clusters:** Can detect clusters of **any shape** (not just circular).
 - **Noise handling:** Clearly **identifies and excludes outliers** from clusters.



The figure above shows a data set with clustering algorithms: K-Means and Hierarchical handling compact, spherical clusters with varying noise tolerance while DBSCAN manages arbitrary-shaped clusters and noise handling.

Key Parameters in DBSCAN

1. eps: This defines the radius of the neighborhood around a data point. If the distance between two points is less than or equal to eps they are considered neighbors. A common method to determine eps is by analyzing the k-distance graph. Choosing the right eps is important:

- If eps is too small most points will be classified as noise.
- If eps is too large clusters may merge and the algorithm may fail to distinguish between them.

2. MinPts: This is the minimum number of points required within the **eps** radius to form a dense region. A general rule of thumb is to set $\text{MinPts} \geq D+1$ where **D** is the number of dimensions in the dataset.

- For most cases a minimum value of **MinPts = 3** is recommended.

How Does DBSCAN Work

- DBSCAN works by categorizing data points into three types:
- Core points which have a sufficient number of neighbors within a specified radius (ϵ)
- Border points which are near core points but lack enough neighbors to be core points themselves
- Noise points which do not belong to any cluster.
- By iteratively expanding clusters from core points and connecting density-reachable points, DBSCAN forms clusters without relying on rigid assumptions about their shape or size.

MinPts = 4
eps = 1 unit

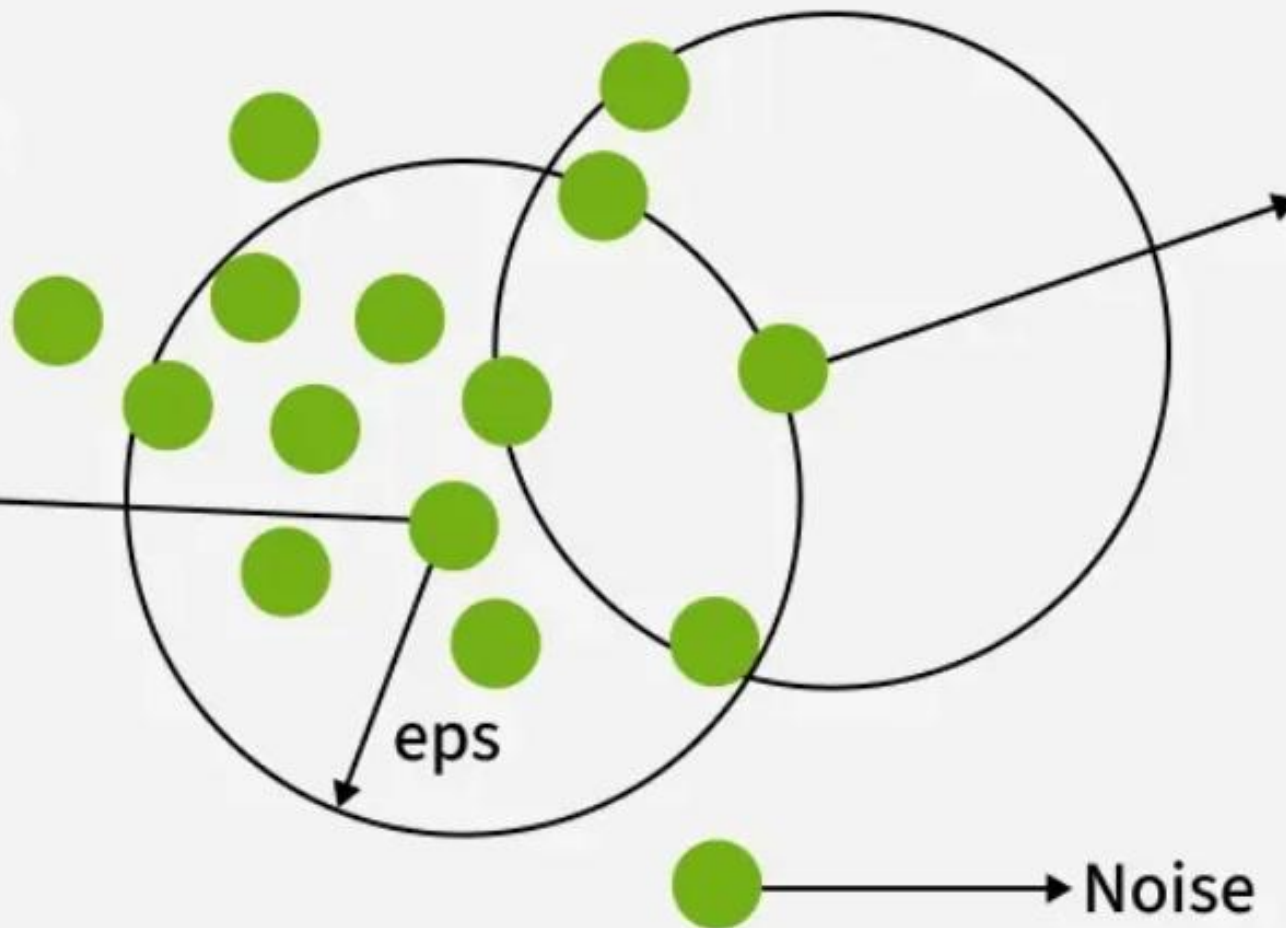
Core Point

Border Point

eps

Noise

DBSCAN ALgorithm



Steps in the DBSCAN Algorithm

- **Identify Core Points:** For each point in the dataset count the number of points within its ϵ neighborhood. If the count meets or exceeds MinPts mark the point as a core point.
- **Form Clusters:** For each core point that is not already assigned to a cluster create a new cluster. Recursively find all density-connected points i.e points within the ϵ radius of the core point and add them to the cluster.
- **Density Connectivity:** Two points a and b are density-connected if there exists a chain of points where each point is within the ϵ radius of the next and at least one point in the chain is a core point. This chaining process ensures that all points in a cluster are connected through a series of dense regions.
- **Label Noise Points:** After processing all points any point that does not belong to a cluster is labeled as noise.

Statistical-based clustering (EM algorithm)

Expectation-Maximization Algorithm

- The Expectation-Maximization (EM) algorithm is a powerful iterative optimization technique used to estimate unknown parameters in probabilistic models, particularly when the data is incomplete, noisy or contains hidden (latent) variables. It works in two steps:
- **E-step (Expectation Step):** Using the current parameter estimates, the algorithm calculates the expected values of the missing or hidden variables. Essentially, it assigns probabilities or "responsibilities" to different hidden outcomes given the observed data.
- **M-step (Maximization Step):** With these updated expectations from the E-step, the algorithm then re-estimates the model parameters by maximizing the expected log-likelihood. This improves how well the model explains the observed data.
- ***These two steps are repeated until convergence, which typically means that:***
 - The parameter values stop changing significantly, or
 - The log-likelihood improves only by a negligible amount.

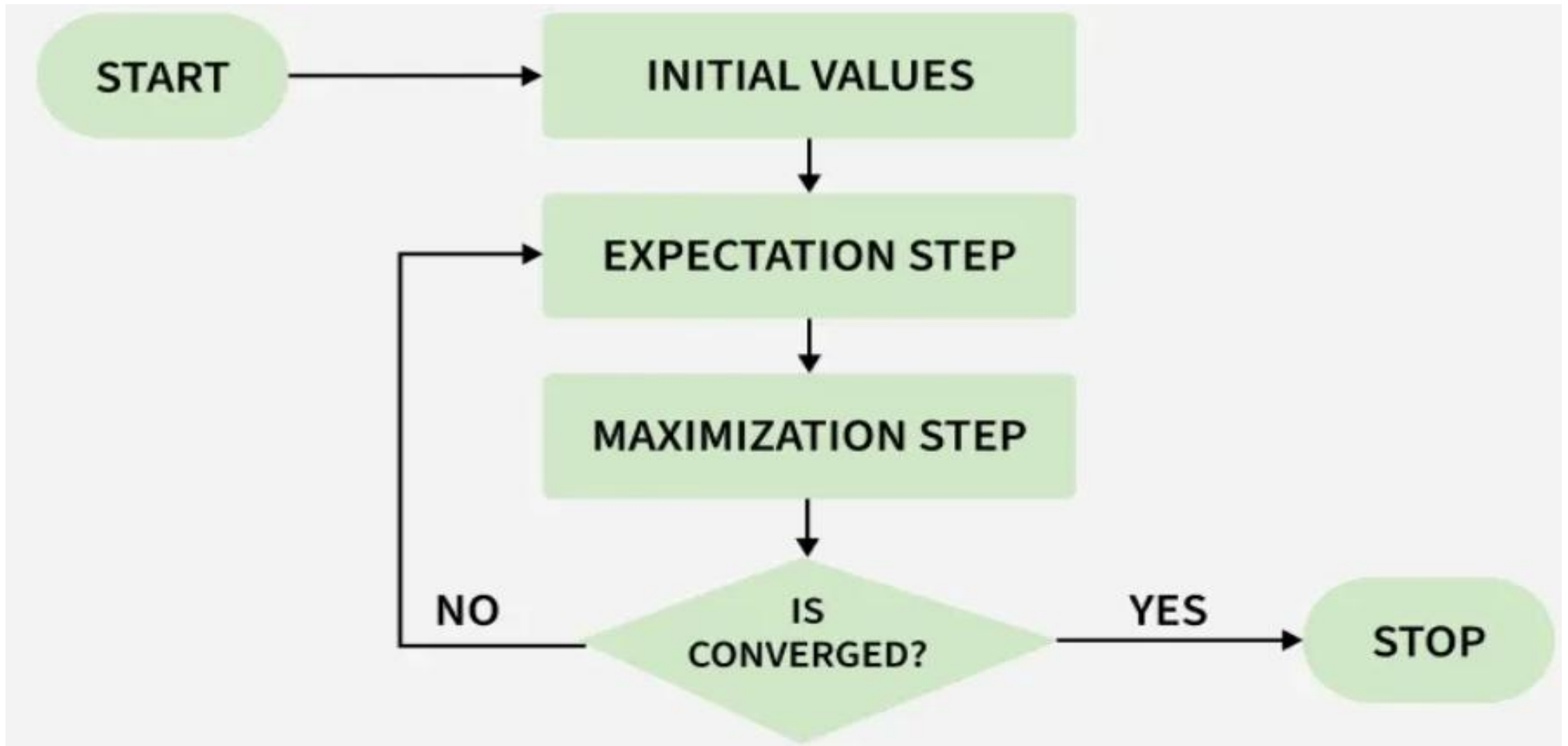


Expectation and Maximization in EM Algorithm

Key Terms in Expectation-Maximization (EM) Algorithm

- **Latent Variables:** Variables that are not directly observed but are inferred from the data. They represent hidden structure (e.g., cluster assignments in Gaussian Mixture Models).
- **Likelihood:** The probability of the observed data given a set of model parameters. EM aims to find parameter values that maximize this likelihood.
- **Log-Likelihood:** The natural logarithm of the likelihood function. It simplifies calculations (turning products into sums) and is numerically more stable when dealing with very small probabilities.
- **Maximum Likelihood Estimation (MLE):** A statistical approach to estimating parameters by choosing the values that maximize the likelihood of observing the given data. EM extends MLE to cases with hidden or missing variables.
- **Posterior Probability:** In Bayesian inference, this represents the probability of parameters (or latent variables) given the observed data and prior knowledge. In EM, posterior probabilities are used in the E-step to estimate the "responsibility" of each hidden variable.
- **Convergence:** The stopping criterion for the iterative process. EM is said to converge when updates to parameters or improvements in log-likelihood become negligibly small, meaning the algorithm has reached a stable solution.

Working of Expectation-Maximization (EM) Algorithm



1. Initialization: The algorithm starts with initial parameter values and assumes the observed data comes from a specific model.

2. E-Step (Expectation Step):

- Find the missing or hidden data based on the current parameters.
- Calculate the posterior probability of each latent variable based on the observed data.
- Compute the log-likelihood of the observed data using the current parameter estimates.

3. M-Step (Maximization Step):

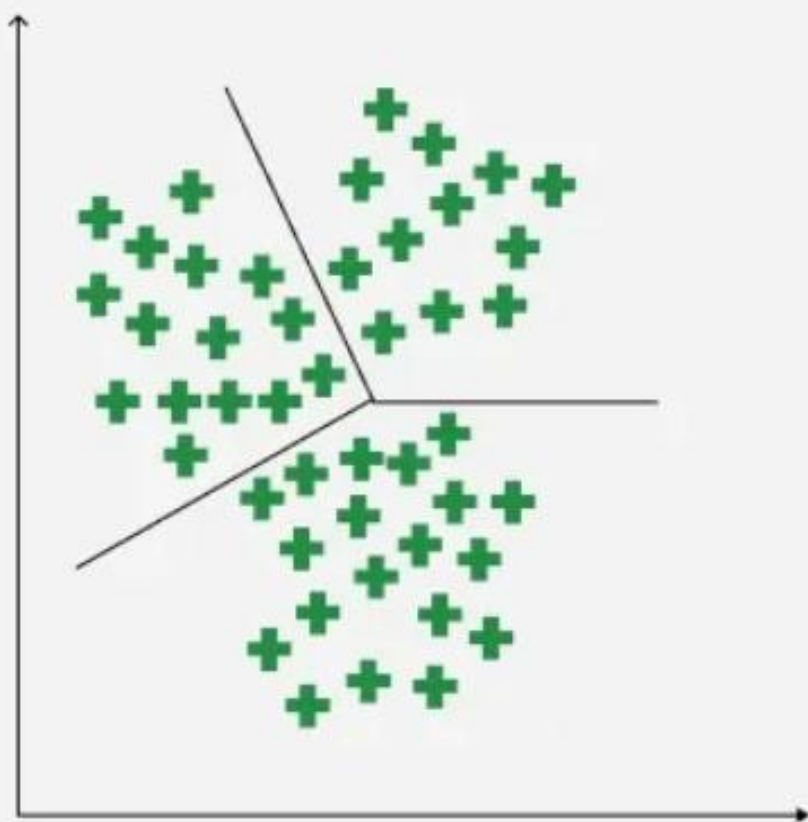
- Update the model parameters by maximize the log-likelihood.
- The better the model the higher this value.

4. Convergence:

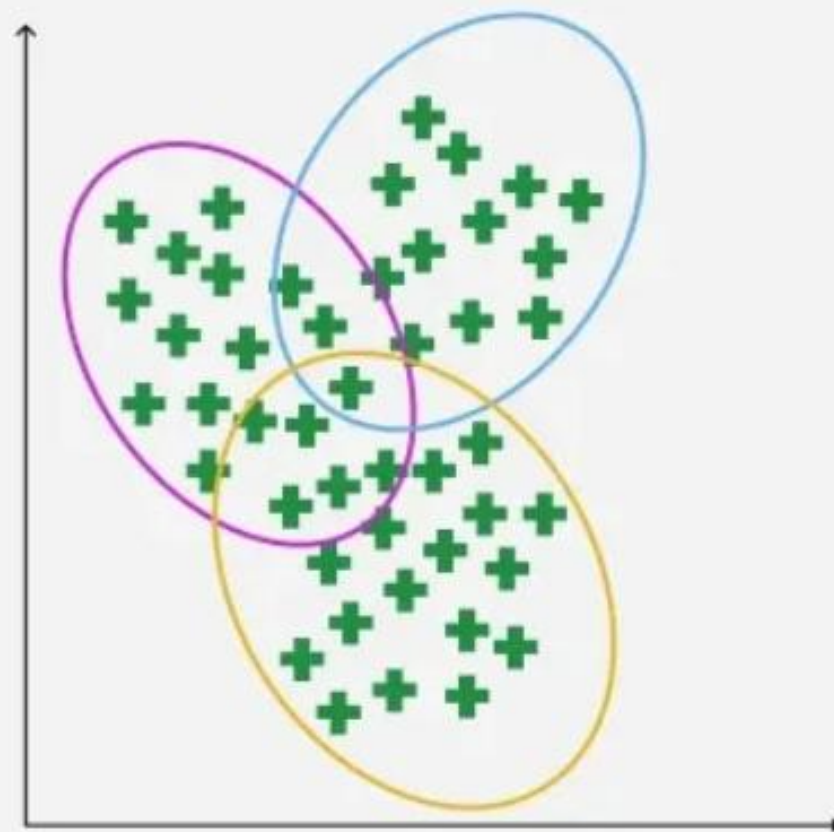
- Check if the model parameters are stable and converging.
- If the changes in log-likelihood or parameters are below a set threshold, stop. If not repeat the E-step and M-step until convergence is reached

Fuzzy Clustering

- Fuzzy clustering allows each data point to belong to multiple clusters with different membership values. Instead of assigning a point to just one group, it captures how strongly a point relates to each cluster.
- Uses membership scores between 0 and 1
- Handles overlapping or unclear cluster boundaries
- More flexible than hard clustering methods
- Useful when data points don't fit neatly into a single group



Hard Clustering



Fuzzy Clustering

Working of Fuzzy Clustering

- Fuzzy clustering assigns each data point a degree of membership for every cluster and updates these values through an iterative process. Here's how it works:

Step 1: Initialize Membership Values Randomly: Each data point is given random membership scores for all clusters. A point can partially belong to multiple clusters.

- For example, for 2 clusters and 4 data points, an initial membership matrix (γ) might look like:

Cluster	(1,3)	(2,5)	(4,8)	(7,9)
1	0.8	0.7	0.2	0.1
2	0.2	0.3	0.8	0.9

Step 2: Compute Cluster Centroids: Centroids are calculated as weighted averages, where weights are membership values raised to the fuzziness parameter m :

$$V_{ij} = \frac{\sum_{k=1}^n \gamma_{ik}^m \cdot x_{kj}}{\sum_{k=1}^n \gamma_{ik}^m}$$

Where:

- γ_{ik} = membership of point k in cluster i .
- m = fuzziness parameter (usually 2).
- x_{kj} = value of feature j for point k .

Step 3: Calculate Distance Between Data Points and Centroids: Compute the Euclidean distance between each point and every centroid to determine proximity, which will be used to update memberships. Example for point (1,3):

$$D_{11} = \sqrt{(1 - 1.568)^2 + (3 - 4.051)^2} = 1.2$$

$$D_{12} = \sqrt{(1 - 5.35)^2 + (3 - 8.215)^2} = 6.79$$

Similarly the distance of all other points is computed from both the centroids.

Step 4: Update Membership Values: Membership values are updated inversely proportional to these distances. Points closer to a centroid get higher membership.

Updated membership γ_{ik} for point k in cluster i is:

Step 5: Repeat Until Convergence: Steps 2–4 are repeated until the membership values stabilize meaning there are no significant changes from one iteration to the next. This indicates that the clustering has reached an optimal state.

Graph-Based Methods

- **Graph clustering** is the process of grouping similar objects (nodes) into clusters within a graph.
- It is used in **data mining** to discover patterns and relationships in large datasets.
- Objects in a graph are grouped based on **similarity** (e.g., shared features or connections).
- Example: In biology, organisms with similar characteristics may belong to the same cluster or species.
- Important parameters include:
 - **Distance between data points**
 - **Density of data points**
- Graph clustering helps in understanding how different data points are **connected or related**.
- It is useful for professionals like **data scientists and retailers** to analyze customer behavior or trends.
- By applying graph clustering, hidden **patterns, structures, and relationships** can be identified.
- It improves decision-making by providing insights from complex data.

Applications of Graph Clustering

1. In the Business World:

- Helps group customers based on **purchasing behavior and preferences**.
- Useful for **customer segmentation** and targeted marketing.
- Classifies products based on **sales and geographic locations**.
- Analyzes the impact of **social media platforms** on business performance.

2. In Biology:

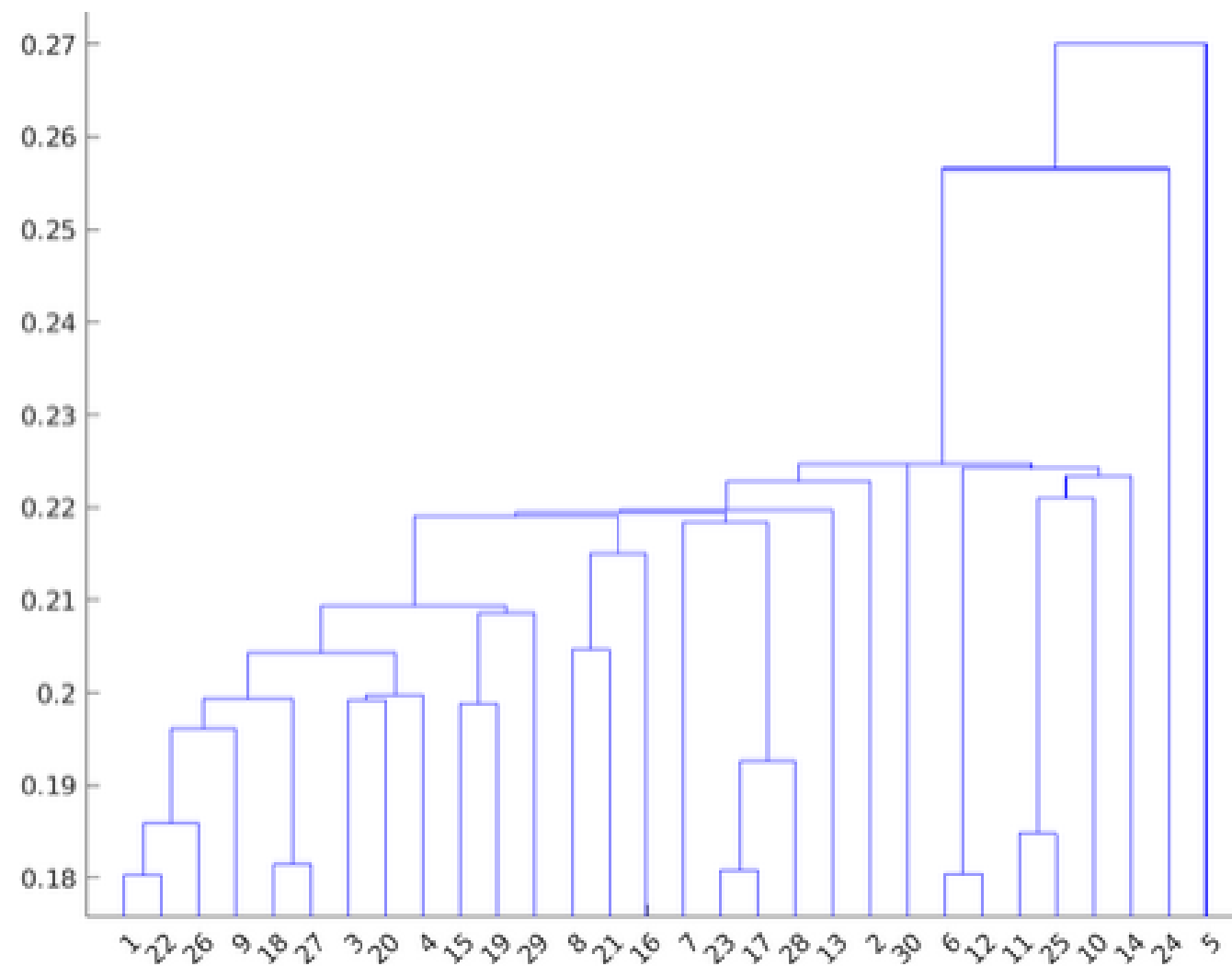
- Used to **classify plants and animals** into groups.
- Helps in studying **genetic similarities** among species.
- Supports identification of **species relationships and common features**.

3. In Geography:

- Helps analyze **forest coverage and population distribution**.
- Groups regions with **similar climatic conditions**.
- Classifies areas based on **rainfall patterns and geographical features**.
- Overall, graph clustering helps extract **useful insights and patterns** across different fields.

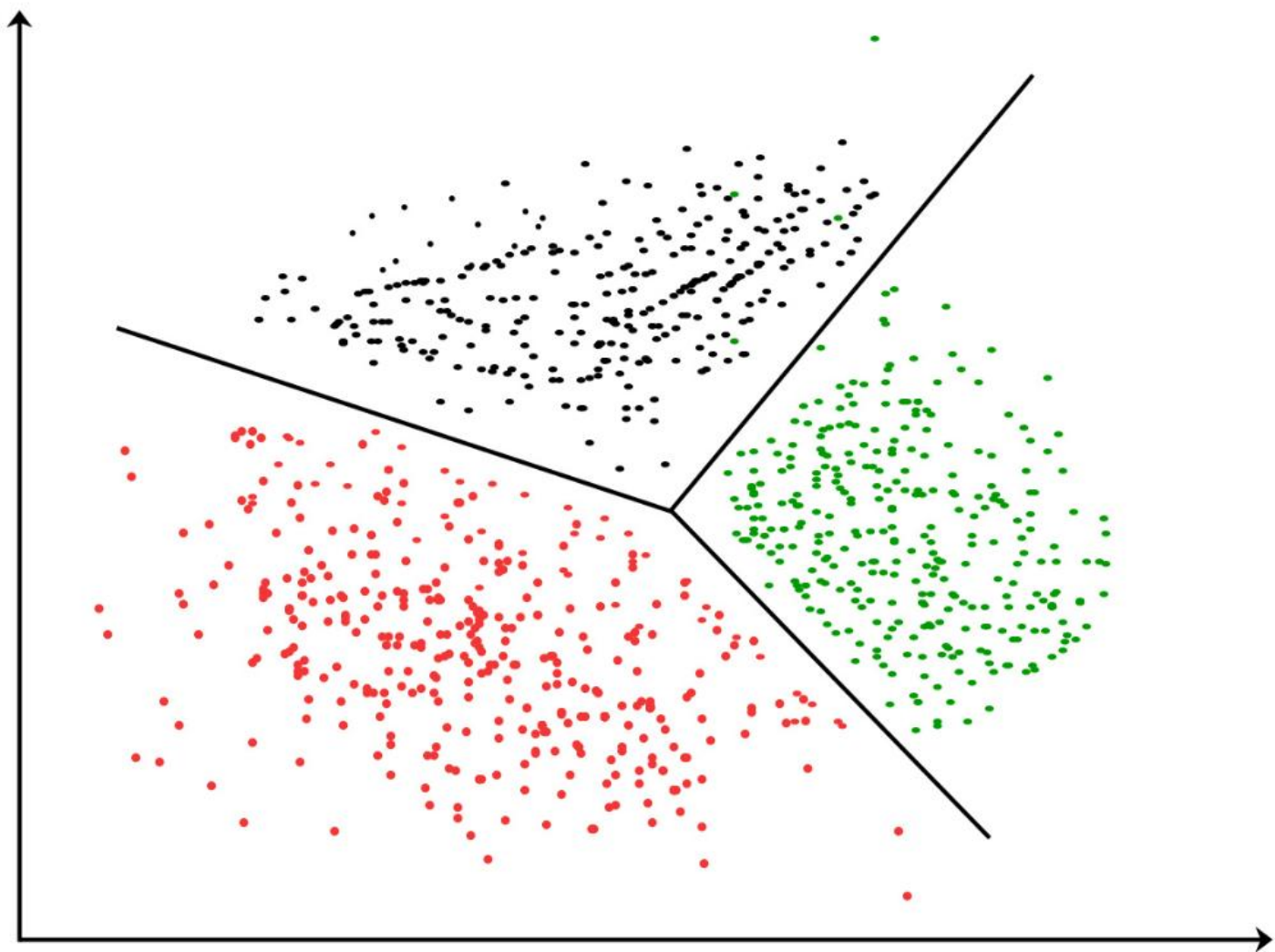
1. Hierarchical graph clustering is a widely used method that organizes data into a **hierarchical structure (tree-like form)**.

- It divides a graph into clusters at different levels of hierarchy.
- There are **two main strategies**:
 - **Divisive Strategy (Top-Down)**:
 - Starts with all data points in **one cluster**.
 - Gradually **splits** clusters into smaller groups at each step.
 - **Agglomerative Strategy (Bottom-Up)**:
 - Starts with each data point as a **separate cluster**.
 - Gradually **merges** clusters until all points form one cluster.
- Each node in the graph represents a **cluster or subgroup**.
- Helps in understanding **data relationships at different levels**.
- **Simple and easy to implement** compared to other methods.
- Provides **effective results** for classifying data into meaningful categories.



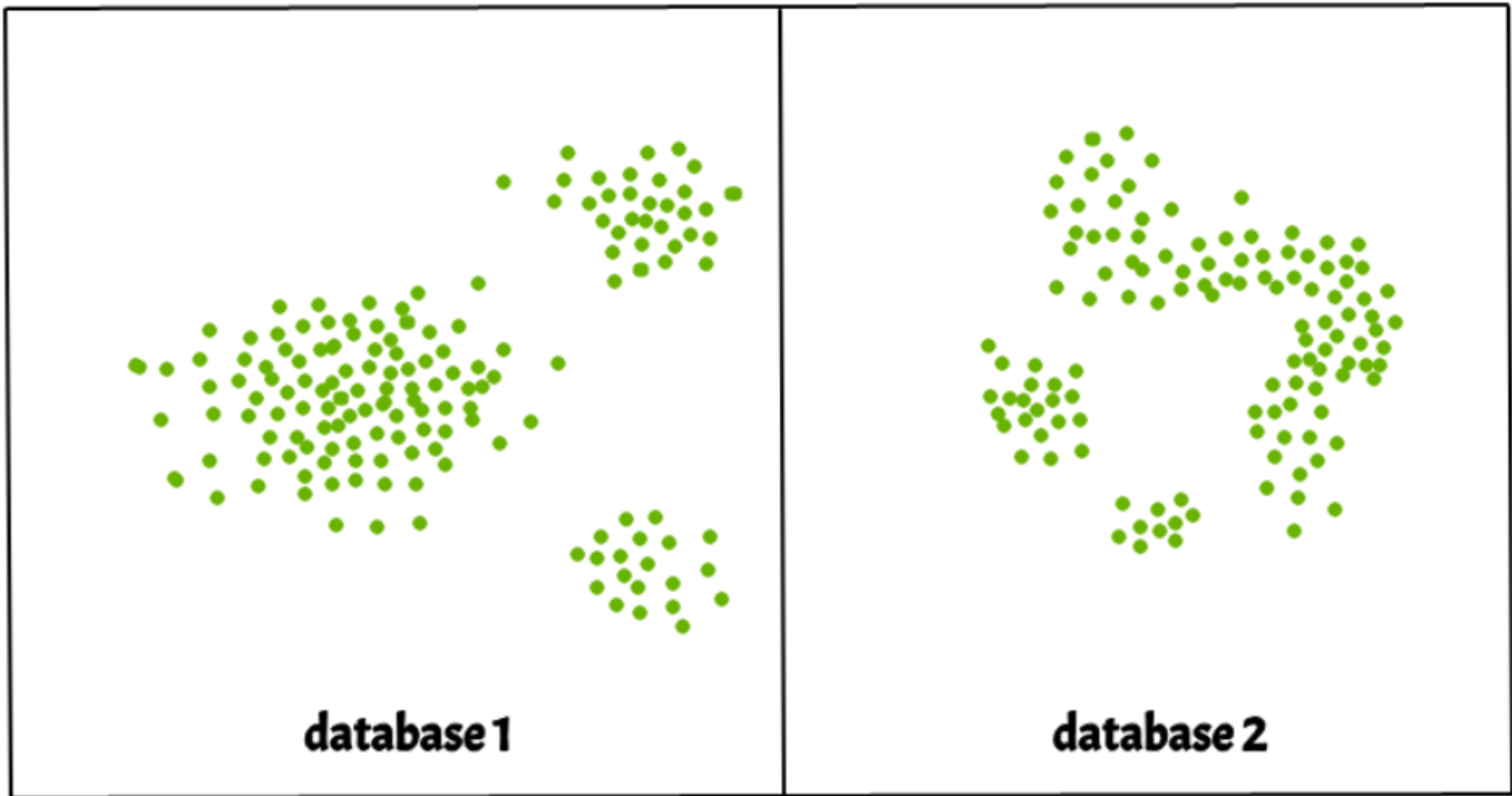
2. K-Means Graph Clustering Method:

- When you use this method, you partition your graph (line chart, bar chart, gauge chart, etc.) into clusters that appear with a k-shape. You can use k-means to compute the centroids and their vector quantities. It takes you easy steps in constructing the K-mean clustering method, which includes:
- You can select K to represent the original centroids.
- Next is to assign points to the nearest centroids from your K-cluster.
- Conduct a re-evaluation of your centroids to ensure they do not change.
- One advantage you can obtain from this method is that it can help you process large data sets.



3. Density-Based Graph Clustering Method:

- Density-based methods work wonders when you want to identify clusters in larger data sets. This is because you can analyze data points based on their density. Two close data points are termed, neighbors. The concept behind this method is density connectivity and density reachability. You will find many data scientists and business persons using the Density-Based Spatial Clustering of Applications with Noise (DBSCAN). Some of the steps you can use in this method include:
- You can begin the clustering process when you find enough data points in your graph.
- Your current data point acts as the starting point.
- Your starting point to the new cluster uses a similar distance which defines its neighborhood.
- You will continue this process in every new cluster until you label all data points.
- An advantage you can find from this graph clustering method is that you can recognize noisy data.



Techniques of Outlier Analysis

- Outliers are data point that is essentially a statistical anomaly, a data point that significantly deviates from other observations in a dataset. Outliers can arise due to measurement errors, natural variation, or rare events, and they can have a disproportionate impact on statistical analyses and machine learning models if not appropriately handled.

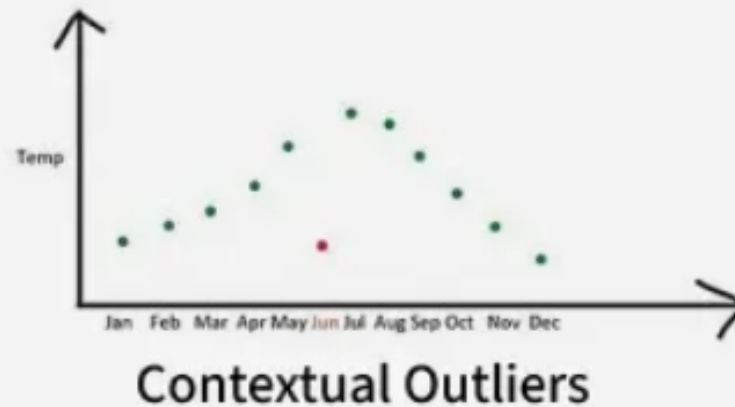
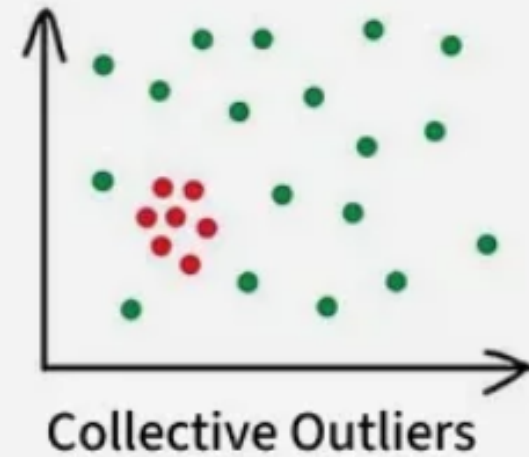
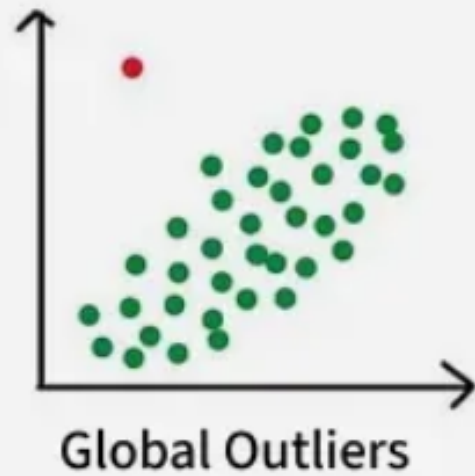
Example: If you have the following dataset of student test scores:

[85, 87, 90, 88, 92, 89, 45]

*The score **45** is an outlier—it's much lower than the others.*

Types of Outliers

Types of Outliers



- **Global Outliers:** Also known as point anomalies, these data points significantly differ from the rest of the dataset.
- **Contextual Outliers:** These are data points that are considered outliers in a specific context. For example, a high temperature may be normal in summer but an outlier in winter.
- **Collective Outliers:** A collection of data points that deviate significantly from the rest of the dataset, even if individual points within the collection are not outliers.

Outlier Detection

- **Outlier detection** is the process of identifying data points that **significantly differ** from the rest of the dataset.
- Outliers are unusual values that can **deviate from normal patterns**.
- Outliers can **distort statistical analysis**:
 - Affect measures like **mean, median, and standard deviation**
 - Lead to **incorrect conclusions and misleading results**

Importance of Outlier Detection

- **Improves Accuracy:**
 - Handling outliers increases the **accuracy and reliability** of models.
- **Fraud Detection:**
 - Helps identify **suspicious or fraudulent activities**, especially in financial data.
- **Maintains Data Quality:**
 - Ensures **clean, consistent, and reliable data** for analysis.
- **Enhances Model Performance:**
 - Improves the performance of **machine learning and statistical models**.
- **Generates Insights:**
 - Outliers may reveal **rare events, new trends, or hidden opportunities**.
- Overall, outlier detection helps in making **better decisions and accurate predictions**.

Methods for Outlier Detection

Standard Deviation Method

Standard Deviation Method is based on the assumption that the data follows a normal distribution. Data point outside of three standard deviations from the mean are considered outliers.

It is commonly used for univariate data analysis where the distribution can be assumed to be approximately normal.

- **Step 1:** Calculate the average and standard deviation of the data set, if applicable.
- **Step 2:** Define the lower and upper bounds for outliers.
- **Step 3:** Identify outliers as data points that fall outside these bounds:

Example: Dataset: [1, 2, 2, 3, 1, 3, 10]. Find an outlier using the Standard Deviation Method.

$$\text{Mean, } \mu = \frac{(1+2+2+3+1+3+10)}{7} = \frac{22}{7} \approx 3.14$$

$$\text{Standard Deviation, } s \approx \sqrt{\frac{(1-3.14)^2 + (2-3.14)^2 + (2-3.14)^2 + (3-3.14)^2 + (1-3.14)^2 + (3-3.14)^2 + (10-3.14)^2}{7}} \approx 2.90$$

Lower and upper bounds for outliers:

- *Lower bound = $3.142857 - 2 \times 2.917$ $3.142857 - 5.834 = -2.691$*
- *Upper bound = $3.142857 + 2 \times 2.917$ $3.142857 + 5.834 = 8.977$*

Any values outside [-2.691, 8.977] is an outlier, thus 10 is an outlier.

So, the data point 10 is identified as an outlier using the Standard Deviation Method.

IQR Method

The Interquartile Range (IQR) method focuses on the spread of the middle 50% of data. It calculates the IQR as the difference between the 75th and 25th percentiles of the data and identifies outliers as those points that fall below 1.5 times the IQR below the 25th percentile or above 1.5 times the IQR above the 75th percentile. This method is robust to outliers and does not assume a normal distribution.

- **Step 1:** Find $Q1$ (25th percentage) and $Q3$ (75th percentage)
- **Step 2:** $IQR = Q3 - Q1$.
- **Step 3:** Find Lower Bound: $Q1 - 1.5 \times IQR$ and Upper Bound Lower Bound: $Q1 - 1.5 \times IQR$

It is suitable for datasets with skewed or non-normal distributions. Useful for identifying outliers in datasets where the spread of the middle 50% of the data is more relevant than the mean and standard deviation.

Example: Dataset $X = \{3, 5, 7, 9, 11, 13, 30\}$, find outlier using the IQR method.

- $Q1$ (25th percentile): Median of first half = Median of $[3, 5, 7] = 5$
- $Q3$ (75th percentile): Median of second half = Median of $[11, 13, 30] = \mathbf{13}$

$$IQR = Q3 - Q1 = 13 - 5 = 8$$

$$\text{Lower Bound: } Q1 - 1.5 \times IQR = 5 - 1.5 \times 8 = 5 - 12 = -7$$

$$\text{Upper Bound: } Q3 + 1.5 \times IQR = 13 + 1.5 \times 8 = 13 + 12 = 25$$

Therefore the interval is -7 to 25. 30 lies outside the interval, therefore is an outlier.

Z-Score Method

The [Z-score method](#) calculates the number of standard deviations each data point is from the mean. A Z-score threshold is set, commonly 3, and any data point with a Z-score exceeding this threshold is considered an outlier. This method assumes a normal distribution and is sensitive to extreme values in small datasets.

- **Step 1:** Calculate the mean.
- **Step 2:** Compute Standard Deviation
- **Step 3:** Calculate z-scores
- **Step 4:** Apply Threshold Rule: Mild outlier: $|Z| > 2$ and Extreme outlier: $|Z| > 3$

Suitable for datasets with large sample sizes and where the underlying distribution of the data can be reasonably approximated by a normal distribution.

Example: $X=\{4,5,5,6,7,8,20\}$, find outlier using the Z-score method.

$$\text{Mean, } \bar{X} = \frac{4+5+5+6+7+8+20}{7} = \frac{55}{7} \approx 7.86$$

$$\text{Standard Deviation, } s \approx \sqrt{\frac{(4-7.86)^2 + (5-7.86)^2 \times 2 + (6-7.86)^2 + (7-7.86)^2 + (8-7.86)^2 + (20-7.86)^2}{6}} \approx 5.36$$

$$\text{Z-scores: } Z_i = \frac{X_i - 7.86}{5.36}$$

Z-score for all data points: 4: -0.72, 5: -0.53, 5: -0.53, 6: -0.35, 7: -0.16, 8: 0.03, 20: **2.26**

Threshold $Z > 2 \Rightarrow 20$ is an outlier since $|2.26| > 2$